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YouTube:

Opened loop	Closed loop
Square-case_1	forward-backward-slow
Square-Case_2	forward-backward-fast
	turn-200 speed
	turn-1000 speed
	square-1
	square-2

GitHub: https://github.com/NoXxChapter/SIT122---Robotics-Studio.git

- Where did the robot end up? Compare the position errors in a table.

The robot's end position usually scatters, lay around the but not back in the original position.

Comparison Table of Open Loop

Start position (cm)	End position (cm)
0	12
0	11
0	12
0	14

The robot's end position usually scatters, lay around the but not back in the original position.

Comparison Table of Close Loop

Start position (cm)	End position (cm)
0	6
0	5
0	8
0	5

- Compare the results from the Open Loop Square Test vs Encoder Motor Square Test in a table. Which one is better? Why?

The return position from both loop case have a significant difference between each one. As we can observe that the close loop have a better result. Which have many reasons, but first must be the wait time allow the inertia to rest to almost zero, greatly reduced the error. But the major must be that with close loop, we measure a lot of information to calculate then execute. Which objectively superior than open loop, when it only blindly run without measure. And there are still numerous

reasons behind but those two maybe the highlight ones gave us a good explanation of why close loop is better than open loop.

What are some of the possible reasons that the robot doesn't end up perfectly at the starting location?

Inertia, floor surface (friction), dust, sand, rock, low/high battery,…

- How did the floor surface affect the positioning error? How do you explain th difference?

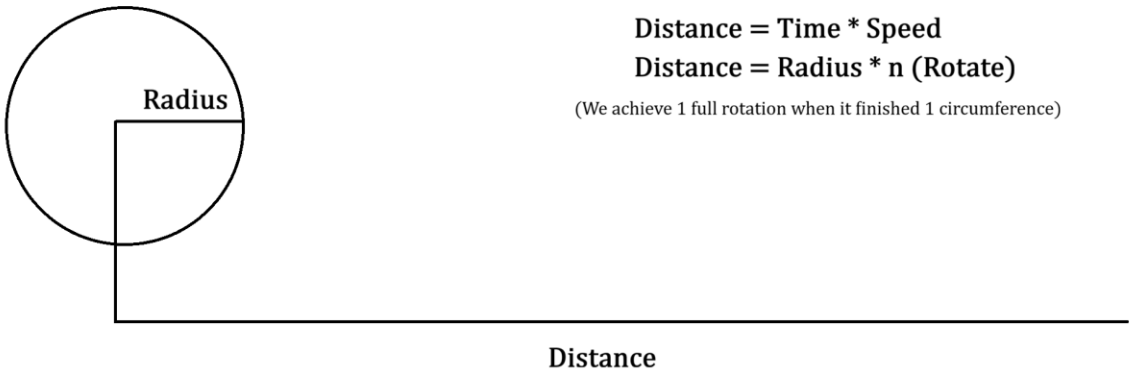
The reason why the floor surface affects the positioning error is first the material. Which with each type can make the interact between the wheels with the surface result in variety. The second were dust, rock, sand. Each can make either the wheel more slippery or roughly.

- What are some of the possible things we can do, to reduce the positioning error?

The easiest solution we can do to achieve the nearly perfect square is run on smooth surface with enough friction to make the wheel stick to floor. Another solution is to change the wheel, use a bigger one which can reduce the affections from external reasons to hardly noticeable.

- For bonus points, explain your math for Encoder Motor Straights and Turns.

Thanks to the guide provided by tutor Precise Motion guideline gave me a quick understand of these measure methods. With both case we have to calculate how many rotation per minute does a wheel(s) have to rotate to successfully execute the request.
First with straight line

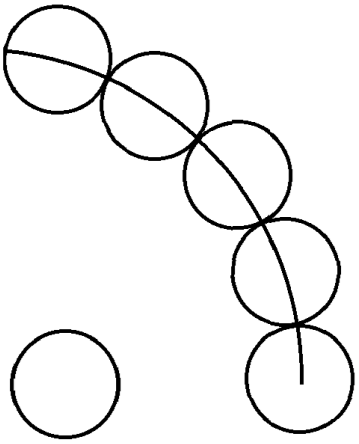


Distance = Time * Speed
Distance = Radius * n (Rotate)
(We achieve 1 full rotation when it finished 1 circumference)

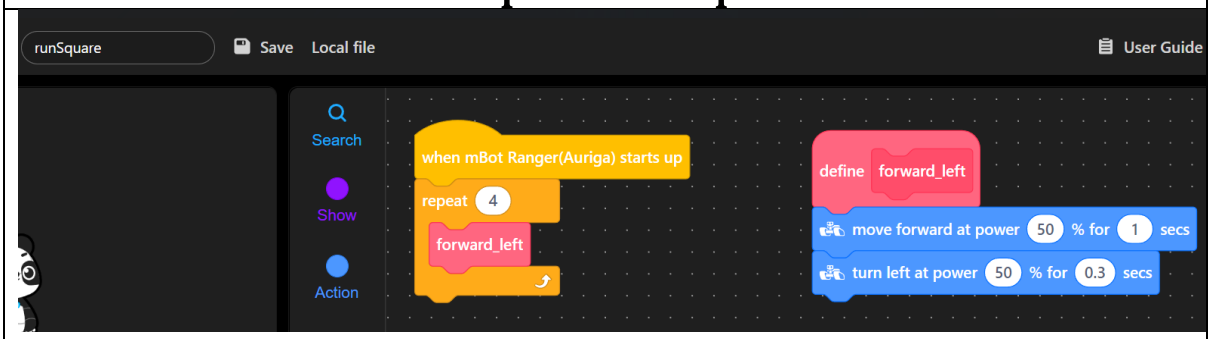
Last is turning

With the same rotation but this case we multiply with the degree need to turn.

Also ONLY move one wheel
- turn right -> move left wheel
- turn left -> move right wheel



Opened Loop



(open loop – square)

Closed Loop

```
define Move distance speed
set radius to 3
set rotate to 360 * distance / 2 * 3.14 * radius
all lights up with color ● for 1 secs
encoder motor onboard motor port1 clockwise rotates by rotate degree at speed of speed r/min
encoder motor onboard motor port2 anticlockwise rotates by rotate degree at speed of speed r/min
wait 3 seconds
set rotate to 360 * 14.5 / 2 * radius
all lights up with color ● for 1 secs
encoder motor onboard motor port1 clockwise rotates by rotate degree at speed of speed r/min
wait 3 seconds

when mBot Ranger(Auriga) starts up
set distance to 50
set speed to 200
forever
  Move distance speed
```

(closed loop – square code)

```
define Move direction distance speed
set radius to 6.5
set rotate to 360 * distance / 2 * 3.14 * radius
if direction = forward then
  all lights up with color ● for 1 secs
  encoder motor onboard motor port1 clockwise rotates by rotate degree at speed of speed r/min
  encoder motor onboard motor port2 anticlockwise rotates by rotate degree at speed of speed r/min
else
  all lights up with color ● for 1 secs
  encoder motor onboard motor port1 anticlockwise rotates by rotate degree at speed of speed r/min
  encoder motor onboard motor port2 clockwise rotates by rotate degree at speed of speed r/min
wait 3 seconds

when mBot Ranger(Auriga) starts up
set distance to 200
set speed to 3000
forever
  Move forward distance speed
  Move backward distance speed
```

(closed loop – forward – backward)